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Comparison of Body Mass Index, Skinfold Thickness, and Bioelectrical Impedance Analysis With Dual-Energy X-Ray Absorptiometry in Hemodialysis Patients

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Abstract

Background: Malnutrition is a consistent finding in hemodialysis (HD) patients and is associated with high mortality. The aim was to compare nutrition status indicators using dual-energy x-ray absorptiometry (DXA) as reference in HD patients. *Methods:* Observational cross-sectional study with 42 patients, 55.8 years (±14.6) old, 60% male, HD 2–3 times per week for ≥3 months. HD ranged from 3 months to 28 years (median, 17.3; interquartile range, 8.73–39.0). We used body mass index (BMI) and fat mass (FM) by skinfold thickness (SFT), bioelectrical impedance analysis (BIA), and DXA. Statistical analyses used Bland-Altman plots, Lin's concordance correlation coefficient, the paired *t*-test, and Pearson or Spearman correlation. $P < .05$ was significant. *Results:* SFT and DXA presented the lowest prevalence of malnutrition (2.4%) and BMI the highest (28.6%). BMI, BIA FM, and SFT FM presented strong positive correlations with DXA FM ($r = 0.915$; $r = 0.976$; $r = 0.910$, $P < .001$, respectively). BIA FM and fat-free mass (FFM) demonstrated substantial agreement with DXA values ($\rho = 0.974$ and 0.960, $P < .001$). Thus, the measurement procedures used, SFT and BIA, underestimated %FM (−4.65% and −2.13%) and overestimated FFM (3.12 kg and 1.0 kg) according to DXA. No differences were found between mean values of BIA FM and DXA ($P = .178$). *Conclusions:* Compared with DXA, BIA was the most appropriate nutrition indicator for measuring body composition. (*Nutr Clin Pract.* 2020;35:1021–1028)

Keywords

bioelectrical impedance analysis; body composition; chronic kidney disease; dual-energy x-ray absorptiometry; fat mass; fat-free mass; hemodialysis; nutrition status

Introduction

Accurate diagnosis of the nutrition status of chronic renal patients is extremely important, since malnutrition is among the highest risks for morbidity and mortality.[1] However, this diagnosis may be complex and not completely accurate, because of frequent alterations in body hydration and the presence of comorbidities and chronic inflammation.[2,3] Studies have investigated the role of body composition through fat mass (FM) and fat-free mass (FFM) and the association with mortality.[4,5] A reduction in lean body mass is an important indicator of malnutrition in patients with chronic kidney disease (CKD),[6] and it was concluded that FM has a protective role in the mortality of hemodialysis (HD) patients.[7]

There is no ideal protocol or single nutrition indicator to evaluate the nutrition status of patients with CKD; however, for use in clinical practice, it is recommended that the indicators are simple, reliable, inexpensive, and easily accessible.[8]

Dual-energy x-ray absorptiometry (DXA) estimates body composition and bone mineral density and is considered as a reference standard indicator by the National Kidney Foundation - Kidney Disease Outcomes Quality Initiative (NKF-KDOQI), which points out that for accurate estimates of body composition and bone mineral density, the use of DXA is preferable to traditional anthropometric indicators.[9] However, because of the need

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for installation and a large space for the equipment, as well as the high cost, its use is restricted in clinical practice.[10]

In this context, body mass index (BMI), skinfold thickness (SFT), and bioelectrical impedance analysis (BIA) are highlighted in the literature as being accessible, practical, and clinically useful indicators to evaluate the nutrition status. In CKD patients, higher BMI values have been related to longer survival,[11,12] and lower values were considered predictors of worse prognosis.[13] The measurement of SFT allows indirect measurement of FM and FFM, and values similar to those of DXA have been found.[14] It has been suggested that higher values of %FM may increase the survival of patients on HD.[15,16] BIA is easy to apply in clinical practice and has been shown to be an appropriate indicator in the estimation of FM and FFM compartments.[17]

It is considered crucial that evaluation of the nutrition status in HD patients is appropriate and accessible. Thus, the aim of the present study was to compare the nutrition status of HD patients through BMI, SFT, and BIA, using DXA as a reference indicator.

Material and Methods

Study Design

This cross-sectional study was conducted between November 2016 and April 2018. The sample included adult and elderly men and women who underwent HD in 2 nephrology clinics. The nonprobabilistic convenience-sample calculation was performed considering a minimum correlation coefficient between the indicators ($r = 0.45$), a bilateral α of 0.05%, and a β of 0.2%, aiming for a study power of 80%, which totaled 36 individuals.[18] The sample considered 170 patients from both clinics; however, 71 presented exclusion criteria, totaling 99 eligible patients. Of these, 57 did not agree to participate, had difficulty in getting to the location of assessments, or were in poor health, resulting in 42 patients evaluated.

The inclusion criteria were being aged ≥19 years, having a diagnosis of CKD, and having undergone HD 2 or 3 times a week for a period of ≥3 months. The exclusion criteria were HD for <3 months; human immunodeficiency virus carriers; individuals with a diagnosis of cancer, advanced heart failure, or chronic lung disease; metal parts in the body; atrophied or amputated limbs; hospitalization; or inability to understand and communicate. Patients with poor health or additional illnesses were excluded to avoid physical overload, as the evaluations were not performed at the nephrology clinics.

The study was approved by the Human Research Ethics Committee, and all participants signed the informed consent form. Personal and demographic data were taken from the patients' medical records or through direct questions to clinic health professionals and patients.

Evaluation of Nutrition Status

All evaluations took place at a single moment, after the HD session, to minimize the interference of hydration and to try to achieve a notional dry weight. The evaluations were performed by a trained researcher, certified as a level 1 anthropometrist by the International Society for the Advancement of Kinanthropometry.

The BMI value (kg/m^2) was calculated using weight (kg) divided by height (m) squared.[19] To measure weight (kg), a Biospace octapolar multifrequency bioimpedance device was used, model InBody 720, with a weight range of 10–250 kg (Biospace, Los Angeles, CA, USA). Height was measured using the portable Exact Height Stadiometer (TBW, São Paulo, Brazil), scale 0.1 cm, maximum length 2.13 m. Patients with a BMI < 23 kg/m^2 were classified as malnourished.[20]

SFTs (bicipital, tricipital, subscapular, and supra-iliac) were measured in the opposite arm to the arteriovenous fistula using a scientific Lange adipometer (Beta Technology Incorporated, Cambridge, MD, USA), according to standard techniques.[21] The average of 3 measurements (mm) was used for the analysis, calculating body density[22] and then %FM.[23] FFM was obtained by subtracting total body weight from the FM. FM values < 10% indicated malnutrition.[20]

BIA body composition measurements were performed using InBody 720 octapolar multifrequency equipment (Biospace, Los Angeles, CA, USA), which contains its own predictor in the accompanying software. This equipment contains 8 electrodes, 4 of which are located on a platform coupled to a scale, on which the patient is positioned in an orthostatic position. The other 4 electrodes are located on 2 levers, each with 2 electrodes that are held by the hands. Patients wore minimal clothing, with bare feet positioned over the platform electrodes. From the analysis of FM provided by the equipment, participants with values < 10% were classified as malnourished.[20]

Reference Method—Dual-Energy X-Ray Absorptiometry

Body composition using DXA was determined with GE Lunar Prodigy Advance equipment and ENCORE 2004, version 8.10.027 (GE Lunar Corporation, Madison, WI, USA). The equipment was calibrated daily and weekly, according to the manufacturer's recommendations. During the evaluation, participants were positioned in the center of the scanner table in a supine position with hands open, facing downwards, and arms extended along the body, without the use of metal adornments.

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Table 1. Clinical Characteristics of Hemodialysis Patients.

Characteristics	Total sample (n = 42)	Men (n = 25)	Women (n = 17)	*P*
Age, y[a]	55.8 (14.6)	53.5 (14.6)	59.2 (14.3)	.215
Dialysis vintage, mo[b]	17.3 (8.7; 39.0)	23.1 (9.4; 40.6)	13.7 (8.7; 27.9)	.391
Serum albumin level, g/dL	3.6 (0.6)	3.78 (0.6)	3.4 (0.4)	**.023**
Weight, kg	71.2 (16.0)	74.0 (15.0)	67.2 (17.0)	.182
BMI, kg/m^2	26.7 (5.6)	25.7 (4.3)	28.1 (7.1)	.192
FM-SFT, kg	20.8 (9.0)	18.4 (7.8)	24.3 (9.7)	.035
%FM-SFT	28.5 (8.2)	24.1 (6.4)	35.1 (6.0)	**<.001**
FM-BIA, kg	22.9 (11.9)	20.2 (10.4)	26.8 (13.1)	.077
%FM-BIA	31.0 (11.7)	26.4 (10.0)	37.9 (10.9)	**.001**
FM-DXA, kg	23.5 (11.6)	20.8 (10.5)	27.3 (12.4)	.074
%FM-DXA	33.2 (11.7)	28.2 (9.9)	40.5 (10.3)	**<.001**
FFM-SFT, kg	50.4 (10.4)	55.6 (8.6)	42.9 (8.0)	**<.001**
FFM-BIA, kg	48.3 (10.2)	53.7 (8.7)	40.4 (6.4)	**<.001**
FFM-DXA, kg	47.3 (9.7)	52.8 (7.6)	39.2 (6.2)	**<.001**

Values in bold indicate $P \leq .05$.
BIA, bioelectrical impedance analysis; BMI, body mass index; DXA, dual x-ray absorptiometry; FFM, fat-free mass; FM, fat mass; SFT, skinfold thickness.
[a]Mean (standard deviation)—Student *t*-test for independent samples.
[b]Median (interquartile range)—Mann-Whitney.

Statistical Analysis

Statistical analysis was performed using Stata 13.0, Data Analysis and Statistical Software (version 13 for Windows—Stata Corporation, College Station, TX, USA). Data are described as mean and standard deviation or median and interquartile range, according to the Shapiro-Wilk normality criteria. χ^2, Student *t*, and Mann-Whitney tests were applied to evaluate differences in body composition between sexes.

Bland-Altman plots, Lin's concordance correlation coefficient (CCC), and the paired *t*-test were used to compare the FM and FFM values obtained by SFT and BIA with those by DXA. The correlations were classified according to the cutoff points proposed by McBride[24] (ρ: >0.99 almost perfect; 0.95–0.99 substantial; 0.90–0.95 moderate; <0.90 poor). Relationships were verified by Pearson or Spearman correlations, and the following interpretation was adopted: *r* from 0 to 0.29 (weak correlation), *r* from 0.30 to 0.69 (moderate correlation), and *r* from 0.70 to 1.0 (strong correlation)[25]. A significance level of 5% ($P < .05$) was considered.

Results

The clinical characteristics of the patients are described in Table 1. Patients presented a mean Kt/V of 1.25 (±0.64), and there was no significant difference between male and female values. Of the total patients, 74% underwent HD 3 times a week, and 10% had undergone renal transplantation but returned to treatment. The main causes of CKD were systemic arterial hypertension (SAH) associated with diabetes mellitus (DM) (40.5%), isolated SAH (23.8%), undetermined cause or other causes (21.4%), isolated DM (9.6%), glomerulonephritis (2.38%), or polycystic kidney disease (2.38%). SAH was the most prevalent comorbidity in the sample (85.7%). The prevalence of malnutrition by BMI was 28.6%, BIA 4.8%, and 2.4% for SFT and DXA.

Although the sample included adults and older adults, no significant differences were found in the estimate of FM (kg) or FFM (kg) evaluated by DXA ($P = .156$ and $P = .721$, respectively). In addition, there were no significant differences between the time of HD (cutoff = median) and FM (kg) evaluated by DXA ($P = .472$).

Patients diagnosed as being nourished by BMI (≥23 kg/m^2) presented significantly higher values of FM (kg) by DXA ($P < .001$) than those classified as malnourished. However, the FFM (kg) by DXA demonstrated no differences between groups ($P = .083$).

According to the analysis of mean differences, it was observed that the measurement procedures used, SFT and BIA, underestimated the FM and overestimated the FFM in relation to DXA (Figure 1). Regarding the underestimation of FM by SFT, it was noted that this underestimation increased as the FM values estimated by DXA increased (Figure 1A and 1B). The overestimation of FFM (kg) by BIA increased as DXA FFM (kg) values increased (Figure 1F). No significant differences were found between the mean FM (kg) values estimated by BIA and DXA.

Based on the results attributed to the residuals, 95.2% (n = 40) of the patients presented an error of estimation of %FM by SFT of between −18.6 kg and 9.3 kg. When this error margin was reduced (−5% and 5%), 42.9% (n = 18) of patients were in this error range (data not shown).

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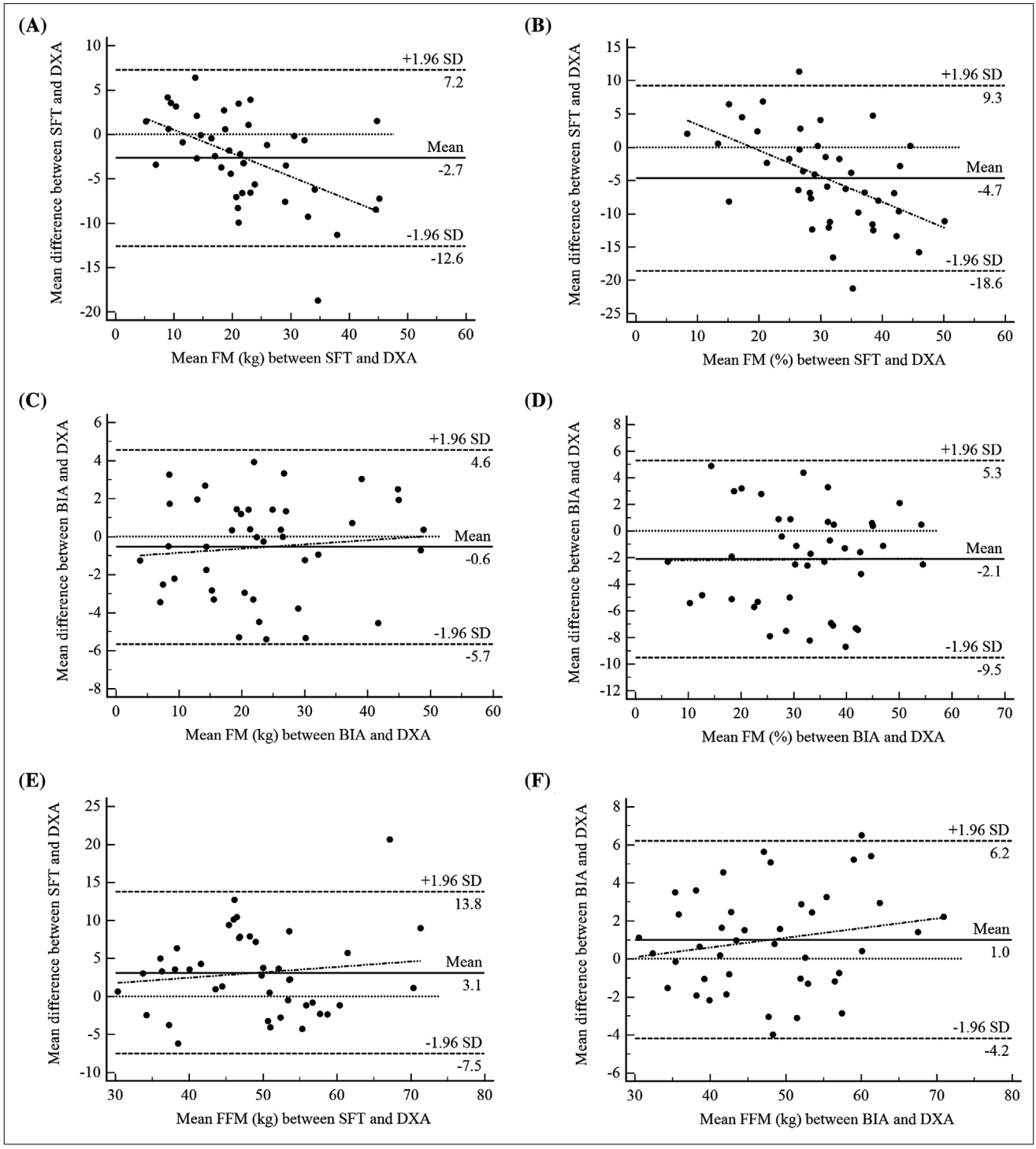

Figure 1. Bland-Altman plots for fat mass (FM) (kg) measured by skinfold thickness (SFT) and dual-energy x-ray absorptiometry (DXA) (A), %FM measured by SFT and DXA (B), FM (kg) measured by bioelectrical impedance analysis (BIA) and DXA (C), %FM measured by BIA and DXA (D), fat-free mass (FFM) (kg) measured by SFT and DXA (E), and FFM (kg) measured by BIA and DXA (F).

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Table 2. Lin's CCC Between FM and FFM by SFT and BIA With Dual X-Ray Absorptiometry.

Variables	CCC (95% CI)	P^a
SFT		
FM, kg	0.853 (0.781–0.924)	<.001
%FM	0.679 (0.544–0.813)	<.001
FFM, kg	0.814 (0.715–0.913)	<.001
BIA		
FM, kg	0.974 (0.959–0.990)	<.001
%FM	0.932 (0.893–0.972)	<.001
FFM, kg	0.960 (0.936–0.984)	<.001

Values in bold indicate $P \leq .05$.
BIA, bioelectrical impedance analysis; CCC, Lin's concordance correlation coefficient; FFM, fat-free mass; FM, fat mass; SFT, skinfold thickness.
[a]Paired *t*-test.

Regarding the results by BIA, it was observed that all patients (n = 42) presented an error of estimation of %FM between −9.5 kg and 5.3 kg. When this error margin was reduced (−5% and 5%), 66.7% (n = 28) of patients were in this error range (data not shown).

BIA met the criterion for substantial agreement for FM (kg) and FFM (kg) values with DXA. For estimation of all components, SFT showed poor agreement with DXA values (Table 2). The FM (kg) obtained by SFT and BIA demonstrated a strong correlation with FM (kg) estimated by DXA ($r = 0.910$, $P < .001$; $r = 0.976$, $P < .001$, respectively). BMI also presented a strong positive correlation with the FM (kg) estimated by DXA ($r = 0.915$, $P < .001$).

Discussion

Although DXA is considered a reference technique for the assessment of body composition,[10,14] it is not routinely available in clinics. Thus, DXA was defined as a reference in the present study for the comparison of nutrition status evaluation to identify easily accessible indicators available for clinical practice.

Depending on the indicator used, the prevalence of malnutrition in HD patients can vary from 23% to 76%.[26] In the present study, we used methods that evaluate body composition using FM (SFT, BIA, and DXA), which presented a low prevalence of malnutrition, justified by the cutoff point (FM < 10%).[20] The International Society for Renal Nutrition and Metabolism recommends that patients with CKD present values above 10%. However, a value below 15% was considered a risk factor for mortality,[12] and a value below 12% led to a risk ratio for mortality that was 4 times higher compared with a group with %FM between 24 and 36.[15] Assuming that detectable alterations in body composition may be developed late in the evolution of malnutrition,[27] a value of FM < 10% established as part of the diagnosis of malnutrition could be questionable.

The mean BMI value of the sample indicated normal weight according to the specific recommendation for CKD.[20] Higher BMI values are independent predictors of better survival in HD patients[28,29] and a cohort study found that a 1 kg/m^2 increase in BMI reduced the risk of mortality.[30] In the present study, BMI was positively correlated with FM (kg) assessed by DXA, a finding consistent with another study.[7] In addition, supporting the study by Donadio et al (2008)[10], patients with a higher BMI presented significantly higher values of FM (kg) estimated by DXA. Some studies[15,31] have assumed that the protective effect of BMI on mortality in HD patients is related to the presence of FM.

The mean FM value evaluated by DXA for the population of the current study is similar to that found in a UK study with 53 adult HD patients, with mean FM values of 21.83 kg (±10.5) and 31.3% (±10.6). FFM values (kg) were also similar (45.69 ± 10.1 kg).[32] Bross et al (2010),[33] in a study with HD patients in California, also found similar values of %FM for their sample (n = 118) of females and males (28.9 ± 10.1; 35.3 ± 8.5; 24.1 ± 8.5, respectively).

The SFT measurement is regularly used because of its practicality and relatively low cost.[8] In agreement with a previous study,[14] there was a significant difference in FM (kg) measured by SFT between sexes. In our study, we found poor agreement between DXA and SFT estimations for all components. However, in the study by Kamimura et al (2003),[14] FM (kg) estimated by SFT showed a strong intraclass correlation coefficient (ICC) with the DXA value. Since ICC and CCC are scaled agreement indices depending on the measurement range, and despite the result in our study, we highlight the relevance of this indicator in clinical practice.

In the present study, SFT presented a strong correlation with FM (kg) assessed by DXA. A similar trend was observed in a study in Mexico[34] ($r = 0.925$) with 20 HD patients (mean age 51.9 ± 19.3 years) and in Poland[35] ($r = 0.868$; $P < .001$) with 48 HD patients (mean age 59.8 ± 15.5 years). However, in our analyses, SFT underestimated FM values and overestimated FFM values when compared with DXA. This has been identified in other studies that found overestimation of FM values by DXA compared with other methods to evaluate body composition.[36,37]

BIA is considered a relatively simple indicator and has been validated in patients with chronic diseases.[38,39] In our study, the body composition estimation assessed by BIA showed a moderate to substantial agreement with estimations by DXA. Kamimura et al (2003)[21] observed this trend and reported a good agreement and strong ICC (r = 0.91) between %FM values assessed by BIA and DXA. In the same study, when analyzed by sex, BIA estimated a lower %FM content for men and higher for women ($P < .001$), which coincides with the present study.

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According to our analysis and compared with DXA, BIA underestimated the %FM values and overestimated FFM. BIA calculates FFM from total body water, based on the assumption that 73.2% of the FFM is composed of water.[40,41] As HD patients present disorders in hydration status,[42] leading to higher volume of total body water, the estimated values found by BIA in relation to the FFM may be justified. In the study of Santos et al (2010)[37] with elite athletes, there was a tendency for DXA to overestimate FM and %FM with increased hydration and underestimate these values with decreased hydration in the FFM compartment. On the other hand, DXA overestimated FFM when hydration decreased and underestimated it when hydration increased. Evidence suggests slight errors in the estimation of fat with changes in hydration,[43] which may be attributed to the fact that, based on the x-ray attenuation properties, water appears to have a fat value of ≈8%.[44,45]

DXA showed good agreement with FM estimated by BIA according to the Bland-Altman analysis, with a mean difference of -0.39 ± 3.3 kg and agreement limits of -6.9 kg to 6.1 kg in all patients.[14] In our study, the mean difference was similar, but the limits of agreement were smaller. Despite the large limits of agreement (+10%),[46] BIA could be of great help in clinical practice for monitoring body-fluid volumes and nutrition markers, such as muscle mass and intracellular volume.[47] It is noteworthy that BIA related patients with a high fat index with longer survival, confirming the already clarified "reverse epidemiology" seen in dialysis, which relates overweight as a positive influence in the survival of HD patients.[48]

It is well known that the aging process is characterized by a progressive decline in FFM, particularly after 45 years of age.[49] Aging-related alterations can still be aggravated when a catabolic condition overlaps, such as in CKD and dialysis patients.[50] However, in the present study, no differences were found in the body composition, evaluated by DXA, of adults and older adults. Furthermore, there was no significant difference between time on dialysis and FM (kg) evaluated by DXA, although it is important to highlight that 74% of the patients had started HD <3 years prior to the study. These findings can be supported by the study by Rosenberger et al (2012),[51] which associated a higher frequency of malnutrition in young patients with longer HD time, and the study by Alvarenga et al (2016),[52] which presented a worsening nutrition status in the group who had undergone HD for 3 years or more.

This study presents as limitations the convenience sample, the use of the internal equation of BIA, and the generalization of the sample regarding disease variability. The strengths include a multimethod study using DXA and nutrition indicators to assess the nutrition status of HD patients. Although hydration and edema were not assessed, all evaluations were performed after HD to minimize possible interferences.

Compared with DXA and considering the clinical applicability in the estimation of body composition in HD patients, BIA was demonstrated to be the most appropriate measurement indicator in patients with similar conditions to those of the present study. Moreover, despite the results found, SFT is probably the most accessible indicator, compatible with the reality of the majority of clinics. Evaluation of nutrition status through appropriate and accessible nutrition indicators will allow adequate nutrition support and monitoring, preserving body composition and reducing the risk of malnutrition.

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Statement of Authorship

A. M. de Abreu and E. Wazlawik equally contributed to the conception and design of the research; A. M. de Abreu and L. C. Wilvert contributed to the acquisition of the data; A. M. de Abreu contributed to the analysis of the data; A. M. de Abreu and E. Wazlawik contributed to the interpretation of the data; and A. M. de Abreu drafted the manuscript. All authors critically revised the manuscript, agree to be fully accountable for ensuring the integrity and accuracy of the work, and read and approved the final manuscript.

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